

Sinhala Multimodal Conversational AI Agent for Automated Food Ordering Using Fine-Tuned Large Language Models

Project Overview

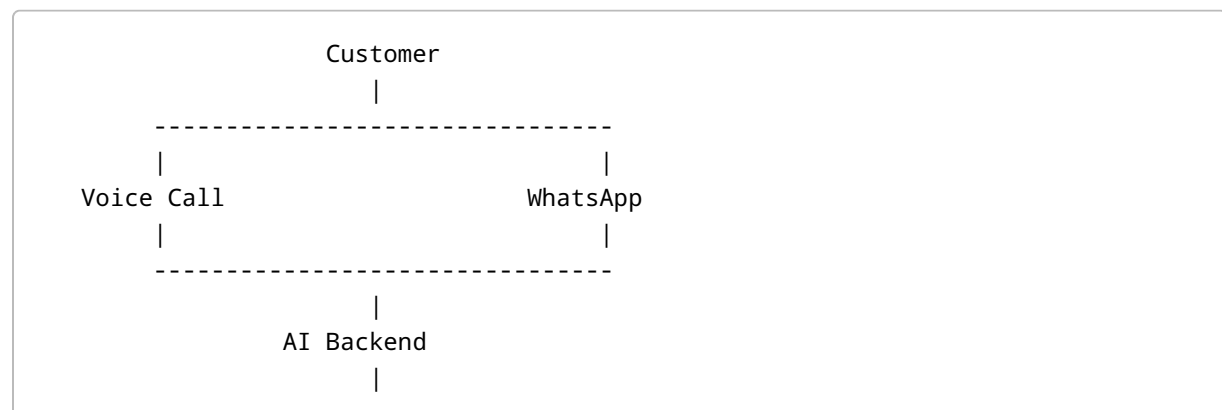
The goal of this project is to develop a Sinhala-speaking AI-powered food ordering assistant that operates through both voice calls and WhatsApp messaging. The system should be capable of understanding customer requests, asking follow-up questions, confirming orders, and forwarding finalized orders to restaurants automatically.

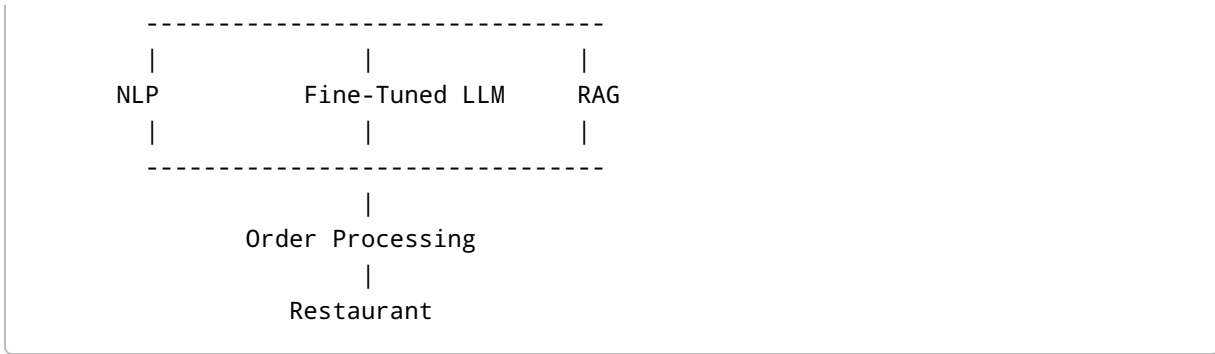
The same AI model should power both communication channels, providing a consistent conversational experience.

Objectives

- Develop a Sinhala conversational AI for food ordering.
- Support both telephone calls and WhatsApp conversations.
- Fine-tune a Large Language Model (LLM) using Sinhala food-ordering conversations.
- Implement Natural Language Processing (NLP) techniques for intent recognition and entity extraction.
- Integrate Retrieval-Augmented Generation (RAG) to access restaurant menus and pricing information.
- Evaluate the effectiveness of fine-tuning compared to baseline models.
- Deploy the system on low-resource infrastructure without requiring a dedicated GPU.

System Architecture





Core Features

Voice Ordering

- Customer calls a phone number.
- AI answers in Sinhala.
- AI understands spoken Sinhala.
- AI collects order details.
- AI confirms order details.
- AI forwards completed orders.

WhatsApp Ordering

- Customer sends messages in Sinhala.
- AI understands requests.
- AI asks clarification questions.
- AI confirms order information.
- AI generates structured orders.

Restaurant Knowledge

- Menu retrieval
- Price retrieval
- Add-ons and customizations
- Availability checking

AI Techniques

1. Natural Language Processing (NLP)

- Text preprocessing

- Text normalization
- Intent classification
- Entity extraction
- Dialogue understanding

Example:

Input:

චීන් කොට්ටු දෙකක් ඔන

Output:

Intent: Food Order
Item: Chicken Kottu
Quantity: 2

2. Transformer-Based Models / LLMs

Potential models:

- Qwen
- Llama
- GPT (for comparison)

Uses:

- Conversation management
- Reasoning
- Context retention
- Order confirmation

3. Fine-Tuning

Train the model using Sinhala food-ordering conversations.

Potential sources:

- WhatsApp conversation exports
- User-generated conversations
- Synthetic conversations generated by LLMs

Research comparison:

- Base model
- Prompt-engineered model
- Fine-tuned model

4. Prompt Engineering

Techniques:

- Zero-shot prompting
- Few-shot prompting
- Chain-of-thought prompting
- System prompt optimization

Example System Prompt:

```
You are a Sinhala food ordering assistant.  
Always confirm item quantities.  
Ask for delivery details.  
Suggest relevant add-ons.
```

5. Retrieval-Augmented Generation (RAG)

Restaurant information stored in:

- Menu database
- Price database
- Availability database

Flow:

```
User Query  
  |  
  v  
RAG Search  
  |  
  v  
Menu Retrieval  
  |
```

Dataset Strategy

Real Dataset

Sources:

- WhatsApp food-ordering conversations
- User-submitted order conversations
- Restaurant customer interactions

Synthetic Dataset

Generate thousands of Sinhala food-ordering conversations covering:

- Formal Sinhala
 - Informal Sinhala
 - Sinhala-English mixed language
 - Slang expressions
 - Spelling mistakes
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Research Questions

1. How effectively can a fine-tuned Sinhala LLM handle food-ordering conversations?
 2. Does fine-tuning improve performance compared to prompt-only approaches?
 3. How accurately can the system identify food-ordering intents and entities?
 4. Can a low-resource Sinhala conversational AI be deployed without dedicated GPU infrastructure?
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Evaluation Metrics

NLP Metrics

- Intent Accuracy
- Precision
- Recall
- F1 Score

Conversation Metrics

- Task Completion Rate
- Average Conversation Turns
- User Satisfaction Score

Speech Metrics

- Word Error Rate (WER)

Model Comparison

Compare:

- Base Model
 - Prompt Engineered Model
 - Fine-Tuned Model
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Deployment Strategy

Training:

- Google Colab GPU

Inference:

- CPU-only deployment
- Quantized models
- Ollama
- llama.cpp

Hosting:

- Local machine
- VPS

Optional voice infrastructure:

- Vapi
 - SIP-based telephony
 - Twilio
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Expected Contributions

- Sinhala conversational AI dataset
 - Fine-tuned Sinhala food-ordering model
 - Multimodal ordering system (Voice + WhatsApp)
 - Evaluation of low-resource language conversational AI
 - Deployable CPU-only architecture
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Expected Outcome

A production-ready Sinhala AI assistant capable of handling food orders through both phone calls and WhatsApp while demonstrating Advanced AI concepts including:

- NLP
- Transformer Models
- Fine-Tuning
- Prompt Engineering
- Retrieval-Augmented Generation (RAG)

This project directly aligns with Advanced AI coursework requirements and demonstrates a practical real-world application of modern AI technologies.